

Formal internal categories and distributors without regularity

Abstract

Cotensor products give a familiar definition of non-Cartesian internal categories, but their existence and stability need not follow from the ambient monoidal structure. We describe a conservative completion of balanced bicomodule cells in which finite paths can serve as both input and output boundaries. Category and distributor actions are then actual cells of a specified calculus. Stable cotensors recover the usual operations. Two examples in abelian groups explain the distinction: finite group-like bases recover enriched categories and distributors in a nonregular ambient category, while a small triangular category admits a nonzero distributor action whose input path has no contextual atomic representative. The latter action is constructed by restriction along a formal functor, on one actual bicomodule.

1 The question

In Aguiar’s formulation, an internal category consists of a comonoid C , a C – C bicomodule A , and associative, unital maps

$$m : A \square_C A \longrightarrow A, \quad u : C \longrightarrow A.$$

Regularity of the monoidal category supplies the cotensors and their compatibility with tensoring [1]. Once regularity is available, categories and distributors fit into the virtual-equipment framework. The question here concerns the operations that remain meaningful when that ambient condition is removed.

The usual virtual treatment of enriched distributors suggests separating operations from represented composites [2]. There is a variance issue, however. Before a bicomodule cotensor exists, the available data are balanced maps *into* a tensor product:

$$N \longrightarrow M_1 \otimes \cdots \otimes M_n.$$

After orienting these as forward maps, they have one input and several outputs. Multiplication and distributor actions have several inputs and one output. We therefore construct a calculus with paths on both sides, while retaining exactly the original balanced maps.

The concrete test is Theorem 6.9. We construct a formal category with carrier $A \cong \mathbb{Z}^5$ and a distributor with carrier $M \cong (\mathbb{Z}/4)^2$. Its action $(A, M) \Longrightarrow M$ is associative, unital, and nonzero, but (A, M) has no contextual representative by any actual bicomodule. An ordinary equalizer exists; its failure to represent the boundary in a tensor context is the point of the example.

Horizontal paths are written in traversal order. Ordinary maps and vertical cell composites are written in functional order, so $ba = b \circ a$; $\odot$ denotes horizontal juxtaposition. A path includes its vertices, and $\emptyset_C$ is the empty path at C . Tensor coherence maps are understood. The general construction uses no braiding. We use $\rightharpoonup$ for bicomodules, distributors, and other horizontal proarrows, following the slashed-arrow convention of [2, Definition 2.1]; the slash is a proarrow marker, not a negation. In double-category diagrams, the same slash marks a horizontal proarrow and a double arrow denotes a cell. In diagrams whose vertices are paths, ordinary arrows denote cells in the merge realization; commutativity asserts equality of their composites. The category and action laws below compare tree composites. Dashed arrows indicate unique factorizations, except where explicitly labelled $\#$.

2 A conservative calculus of paths

2.1 The balanced input

Definition 2.1 (Bicomodules and equivariant maps). Write $M : C \rightharpoonup D$ for a bicomodule with commuting, coassociative, counital coactions $\lambda_M : M \rightarrow C \otimes M$ and $\rho_M : M \rightarrow M \otimes D$. An equivariant map $F : M \rightarrow M'$ over comonoid maps $f : C \rightarrow C'$, $g : D \rightarrow D'$ satisfies

$$\begin{array}{ccc} M & \xrightarrow{F} & M' \\ \lambda_M \downarrow & & \downarrow \lambda_{M'} \\ C \otimes M & \xrightarrow{f \otimes F} & C' \otimes M' \end{array} \quad \begin{array}{ccc} M & \xrightarrow{F} & M' \\ \rho_M \downarrow & & \downarrow \rho_{M'} \\ M \otimes D & \xrightarrow{F \otimes g} & M' \otimes D'. \end{array}$$

Definition 2.2 (Balanced cells). For a path $M_i : C_{i-1} \rightharpoonup C_i$, an output bicomodule $N : D \rightharpoonup E$, and comonoid maps $a : D \rightarrow C_0$, $b : E \rightarrow C_n$, a balanced map $\phi : N \rightarrow T$, where $T = M_1 \otimes \cdots \otimes M_n$, has exterior equivariance

$$\begin{array}{ccc} N & \xrightarrow{\phi} & T \\ \lambda_N \downarrow & & \downarrow \lambda_{M_1} \otimes 1 \\ D \otimes N & \xrightarrow{a \otimes \phi} & C_0 \otimes T \end{array} \quad \begin{array}{ccc} N & \xrightarrow{\phi} & T \\ \rho_N \downarrow & & \downarrow 1 \otimes \rho_{M_n} \\ N \otimes E & \xrightarrow{\phi \otimes b} & T \otimes C_n \end{array} \quad (2.1)$$

and interior balance. Write T_i for T with C_i inserted between M_i and M_{i+1} , and put $r_i = 1 \otimes \rho_{M_i} \otimes 1$ and $\ell_i = 1 \otimes \lambda_{M_{i+1}} \otimes 1$. Balance means that the two routes below have a common composite ψ_i :

$$\begin{array}{ccc} N & \xrightarrow{\phi} & T \\ & \searrow \psi_i & \downarrow \ell_i \downarrow r_i \\ & & T_i. \end{array} \quad (1 \leq i < n). \quad (2.2)$$

For the empty path at C , a balanced map is $\phi : N \rightarrow I$ making

$$\begin{array}{ccc} N & \xrightarrow{\rho_N} & N \otimes E \\ \lambda_N \downarrow & & \downarrow \phi \otimes b \\ D \otimes N & \xrightarrow{a \otimes \phi} & C \end{array}$$

commute. Regard these maps as cells $(M_1, \dots, M_n) \Rightarrow N$:

$$\begin{array}{ccccc} C_0 & \xrightarrow{M_1} & C_1 & \cdots & C_{n-1} & \xrightarrow{M_n} & C_n \\ a^{\text{op}} \downarrow & & & \Downarrow \phi & & & \downarrow b^{\text{op}} \\ D & \xrightarrow{\quad} & \cancel{N} & \xrightarrow{\quad} & E \end{array}$$

The displayed exterior arrows are in the opposite comonoid category.

Proposition 2.3 (The balanced virtual equipment). *For every monoidal $\mathcal{V}$, balanced cells form a virtual equipment*

$$\mathbb{C}(\mathcal{V}) = \text{Mod}(\mathcal{V}^{\text{op}}),$$

with comonoids as objects and opposite comonoid maps as vertical arrows. Its restrictions are corestrictions of coactions, and its units are the regular bicomodules. Here Mod is the monoids-and-bimodules construction, in its virtual-double-category form [2, Definition 2.8].

Proof. Substitution is composition with the tensor of the maps inserted into the factors. Equations (2.1)–(2.2), including the empty case, are preserved by this substitution.

Restrictions retain the underlying object and corestrict its coactions along the actual comonoid maps. The identity map is cartesian, since factorization through it imposes exactly the same equivariance equations. The regular bicomodule C represents $\emptyset_C$, with comparison given by ε_C . Indeed a balanced $\phi : N \rightarrow I$ has the unique equivariant factorization over (a, b)

$$\begin{array}{ccc} N & \xrightarrow{\exists! \bar{\phi}} & C \\ & \searrow \phi & \downarrow \varepsilon_C \\ & & I, \end{array} \quad \bar{\phi} = (a \otimes \phi) \lambda_N = (\phi \otimes b) \rho_N.$$

Coassociativity gives equivariance; applying the counit to either equivariance square forces this formula and proves uniqueness. In a nonempty context, insert the regular factor using the adjacent coaction; balance identifies the two insertion formulas and the counit removes the inserted factor. This proves the contextual unit property. $\square$

2.2 Forests and intrinsic comparisons

Definition 2.4 (The forest category). Let P be the entire globular part of a virtual equipment $\mathbb{D}$, retaining arrows between different objects. Call its original horizontal arrows *atomic*. Let $\text{For}(P)$ be the strict 2-category with finite paths as 1-arrows. A 2-cell $\Gamma \Rightarrow (K_1, \dots, K_m)$ is a subdivision of Γ into m consecutive, possibly empty blocks, together with an original cell from the j th block to K_j . Cut vertices must agree. For an empty output there is only the identity forest on the empty path. Vertical composition substitutes forests and horizontal composition concatenates them. In particular,

$$\text{For}(P)(\Gamma; K) = P(\Gamma; K) \quad \text{for atomic } K.$$

Definition 2.5 (Contextual representations). An original cell $s : \Omega \Rightarrow T$ is *contextually universal* if every original cell α has the unique factorization

$$\begin{array}{ccc} (L, \Omega, R) & \xrightarrow{1_L \odot s \odot 1_R} & (L, T, R) \\ & \searrow \alpha & \downarrow \exists! \bar{\alpha} \\ & & K \end{array} \tag{2.3}$$

for every compatible pair of paths L, R and every atomic K . Equivalently, substitution is a bijection $P(L, T, R; K) \cong P(L, \Omega, R; K)$. Let $S(P)$ be the class of these comparisons, including empty-source ones. A path is *represented* when it has such a comparison to an atom.

Remark 2.6 (Exterior boundaries). The contexts in (2.3) may pass through other objects. Cartesian restriction of the target gives $\mathbb{D}_{f,g}(\Gamma; K) \cong P(\Gamma; K[f, g])$, so this globular test is equivalent to the same test with arbitrary exterior vertical maps.

Definition 2.7 (The local merge calculus). Define $L(P)$ by adjoining to $\text{For}(P)$ an inverse $s^{-1} : T \Rightarrow \Omega$ for each $s \in S(P)$, with the two inverse equations. The equations are closed under both compositions. Equivalently, localize at all $1_L \odot s \odot 1_R$ in the hom-categories. Horizontal composition extends because it takes such instances to such instances.

Theorem 2.8 (Conservativity and detection). *The canonical map*

$$P(\Gamma; K) \longrightarrow L(P)(\Gamma; K) \tag{2.4}$$

is a bijection for every atomic K , compatible with substitution. Moreover, Γ is isomorphic to an atom T in $L(P)$ exactly when Γ is represented by T in the sense of Definition 2.5.

Proof. For an atom K , the substitution presheaf $\Phi_K(\Gamma) = P(\Gamma; K)$ on $\mathbf{For}(P)$ takes contextual comparisons to bijections. It therefore extends to $L(P)$ by making their formal inverses act by inverse bijections. If original cells have the same image, apply their presheaf actions to 1_K ; this recovers the cells and proves injectivity.

For surjectivity, express $z : \Gamma \Longrightarrow K$ as a vertical sequence of forests and inverses of contextual comparisons. Interchange provides such a sequence. Starting with 1_K , read it backwards. At a forest, substitute into the current original cell. At the inverse of $t = 1_L \odot s \odot 1_R$, factor the current cell uniquely as ct , using (2.3); then $(ct)t^{-1} = c$. Each step replaces the localized suffix by an original cell. At the end, z is an original cell. These operations are exactly the extended presheaf action, so the answer is independent of the expression.

Finally, an isomorphism $\Gamma \rightarrow T$ is an original cell by conservativity. Whiskering it by L, R and precomposing into any atomic output gives a bijection. Equation (2.4) identifies this with (2.3), proving detection. The converse holds by the definition of the localization. $\square$

Corollary 2.9 (Universal property). *An interpretation of P in a strict 2-category $\mathcal{T}$ extends uniquely to forests. The resulting G has the unique extension below if and only if it inverts $S(P)$:*

$$\begin{array}{ccc} \mathbf{For}(P) & \xrightarrow{\ell} & L(P) \\ & \searrow G & \downarrow \exists! \bar{G} \\ & & \mathcal{T}. \end{array}$$

Proof. A forest must be sent to the juxtaposition of its constituent cells. An extension across $L(P)$ must send each s^{-1} to $G(s)^{-1}$; these assignments respect exactly the defining relations of Definition 2.7. $\square$

2.3 Vertical boundaries and the two-sided reduct

Lemma 2.10 (Binding comparisons). *For a vertical arrow $f : X \rightarrow X'$, choose local binding presentations $f_* : X \rightrightarrows X'$ and $f^* : X' \rightrightarrows X$ from restrictions of units. Their binding cells yield in $L(P)$ an adjunction with*

$$j_f : \emptyset_X \Longrightarrow f_* f^*, \quad e_f : f^* f_* \Longrightarrow \emptyset_{X'}.$$

Its triangle identities are

$$\begin{array}{ccc} f_* & \xrightarrow{j_f \odot 1} & f_* f^* f_* \\ & \searrow & \downarrow 1 \odot e_f \\ & & f_* \end{array} \quad \begin{array}{ccc} f^* & \xrightarrow{1 \odot j_f} & f^* f_* f^* \\ & \searrow & \downarrow e_f \odot 1 \\ & & f^*. \end{array}$$

The base-change comparisons

$$f_* K g^* \Longrightarrow K[f, g], \quad k_{f,u} : f_* u_* \Longrightarrow (uf)_*, \quad l_{f,u} : u^* f^* \Longrightarrow (uf)^*$$

are contextually universal and hence invertible [2, Theorem 7.16 and Corollary 7.19]. The comparisons k, l are associative and compatible with j, e .

Proof. The triangle identities are the companion and conjoint binding identities, with the represented units identified with empty paths. The first comparison is obtained by adjoining the companion and conjoint binding cells to K and factoring through its restriction. Inserting the other binding cells gives the inverse factorization in every context. The same argument for pasted binding pairs gives k, l . Uniqueness of binding-compatible comparisons proves their associativity and their compatibility with j, e . $\square$

Definition 2.11 (The global merge calculus). For a square with path boundaries

$$\begin{array}{ccc} X & \xrightarrow{\Gamma} & Y \\ f \downarrow & \Downarrow a & \downarrow g \\ X' & \xrightarrow{\Delta} & Y' \end{array}$$

define the merge realization by

$$E(\mathbb{D})_{f,g}(\Gamma; \Delta) = L(P)(\Gamma; f_*\Delta g^*) \quad (2.5)$$

in any such local presentation. More intrinsically, take the disjoint union over the two binding presentations and identify representatives by their unique binding-compatible isomorphisms. Only finite local choices are used in any pasting. Normalize identity vertical maps to empty companion and conjoint paths.

For representatives $\hat{a}, \hat{b}$ in (2.5), horizontal pasting of adjacent squares with sides (f, g) and (g, h) is the outer route of

$$\begin{array}{ccc} \Gamma\Gamma' & \xrightarrow{\hat{a}\hat{\odot}\hat{b}} & f_*\Delta g^*g_*\Delta'h^* \\ & \searrow \hat{a}\hat{\odot}b & \downarrow 1\odot e_g\odot 1 \\ & & f_*\Delta\Delta'h^*. \end{array}$$

If b below a has sides (u, v) and bottom boundary Ξ , vertical pasting is defined by the square

$$\begin{array}{ccc} \Gamma & \xrightarrow{\hat{a}} & f_*\Delta g^* \\ \hat{b}\hat{a} \downarrow & & \downarrow 1\hat{\odot}\hat{b}\odot 1 \\ (uf)_*\Xi(vg)^* & \xleftarrow{k_{f,u}\odot 1\Xi\odot l_{g,v}} & f_*u_*\Xi v^*g^* \end{array}$$

Vertical identities are path identities; the horizontal identity on f is j_f .

Proposition 2.12 (Coherence of the merge calculus). *Definition 2.11 gives a strict double category with paths as horizontal arrows. Its operations are independent of the local binding presentations.*

Proof. The triangle identities of Lemma 2.10 prove the unit laws. Horizontal associativity contracts disjoint counits; vertical associativity is that of k, l . Interchange is the composite-counit square

$$\begin{array}{ccc} v^*g^*g_*v_* & \xrightarrow{l_{g,v}\odot k_{g,v}} & (vg)^*(vg)_* \\ 1\odot e_g\odot 1 \downarrow & & \downarrow e_{vg} \\ v^*v_* & \xrightarrow{e_v} & \emptyset_{Y''}. \end{array}$$

Here $g : Y \rightarrow Y'$ and $v : Y' \rightarrow Y''$. These diagrams commute by the same binding identities, and are invariant under changes of local presentation by uniqueness of the binding-compatible comparisons. $\square$

Theorem 2.13 (Global conservativity and boundary transport). *The inclusion of $\mathbb{D}$ into $E(\mathbb{D})$ is bijective on cells with one atomic output, with arbitrary exterior vertical maps. It preserves substitution and preserves and reflects the original cartesian cells. Companion and conjoint mates give coherent transport of both boundaries.*

Proof. For one atomic output the required bijection is the top edge of

$$\begin{array}{ccc} \mathbb{D}_{f,g}(\Gamma; K) & \xrightarrow{\cong} & \mathbb{E}(\mathbb{D})_{f,g}(\Gamma; K) \\ \cong \downarrow & & \parallel \\ P(\Gamma; K[f, g]) & \xrightarrow{\cong} & L(P)(\Gamma; f_* K g^*). \end{array} \quad (2.6)$$

The left bijection is cartesian factorization; the bottom one uses Theorem 2.8 and the base-change comparison. Binding pairs cancel at internal seams, proving compatibility with substitution. These bijections also preserve and reflect the original cartesian cells. The mate bijections form the triangle

$$\begin{array}{ccc} \mathbb{E}(\mathbb{D})_{f,g}(\Gamma; \Delta) & \xrightarrow{\cong} & L(P)(\Gamma; f_* \Delta g^*) \\ & \searrow \cong & \downarrow \text{mates} \\ & & L(P)(f^* \Gamma g_*; \Delta) \end{array}$$

and supply coherent transport of both boundaries. Their compatibility with pasting follows from the binding identities of Lemma 2.10. $\square$

Definition 2.14 (The two-sided balanced calculus). The *poly-virtual reduct* U_{poly} retains all cells and equalities of this merge calculus and specifies its tree operations: connected, acyclic rectangular pastings along atomic wires, with matching vertical seams. Structural identity wires are contracted, unused through-wire components are excluded, and identity wires and empty squares on vertical maps are also retained as structural operations. Two parallel internal wires form a cycle, so composition along an entire multiwire boundary is a merge operation. Nullary vertices are allowed. The monad and action equations below use only tree substitution; merge operations remain available in the chosen realization for constructing their cells.

Finally set

$$\mathbb{B}(\mathcal{V}) = \mathbb{E}(\mathbb{C}(\mathcal{V}))^\vee, \quad \mathbb{Q}(\mathcal{V}) = U_{\text{poly}} \mathbb{B}(\mathcal{V}). \quad (2.7)$$

Here vertical reversal reverses both the vertical category and the squares, retaining horizontal directions. Thus vertical arrows are now forward comonoid maps.

Corollary 2.15 (Primitive and unary cells). *The calculus $\mathbb{Q}(\mathcal{V})$ preserves and reflects every primitive single-input balanced cell. Its unary squares between actual bicomodules are exactly forward equivariant maps, and it has coherent boundary transport.*

Proof. Reverse the squares in Theorem 2.13. Its single-output bijections become single-input bijections, and its two mate descriptions exchange. The reduct retains all these cells and equalities. $\square$

3 Formal categories and cotensor recovery

Definition 3.1 (Formal internal categories). A *formal internal category* is a comonoid C , an actual endobicomodule $A : C \rightharpoonup C$, and globular cells

$$\mu_A : (A, A) \Longrightarrow A, \quad \eta_A : \emptyset_C \Longrightarrow A$$

in $\mathbb{Q}(\mathcal{V})$, satisfying

$$\begin{array}{ccc} (A, A, A) & \xrightarrow{(\mu_A, 1)} & (A, A) \\ (1, \mu_A) \downarrow & & \downarrow \mu_A \\ (A, A) & \xrightarrow{\mu_A} & A \end{array} \quad \begin{array}{ccc} (A, A) & \xleftarrow{(\eta_A, 1)} & A \xrightarrow{(1, \eta_A)} (A, A) \\ & \searrow \mu_A & \parallel \\ & & A. \end{array} \quad (3.1)$$

Write $\mathbf{A} = (C, A, \mu_A, \eta_A)$.

Definition 3.2 (Formal functors). A formal functor $F = (f, F) : A \rightarrow B$ consists of a comonoid map f and a unary square F over (f, f) , satisfying

$$\begin{array}{ccc} (A, A) & \xrightarrow{\mu_A} & A \\ (F, F) \downarrow & & \downarrow F \\ (B, B) & \xrightarrow{\mu_B} & B \end{array} \quad \begin{array}{ccc} \emptyset_C & \xrightarrow{\eta_A} & A \\ [f] \downarrow & & \downarrow F \\ \emptyset_D & \xrightarrow{\eta_B} & B. \end{array} \quad (3.2)$$

Here C, D are the object comonoids and $[f]$ is the structural empty square on f .

Proposition 3.3 (Composition of formal functors). *Formal internal categories and formal functors form a category $\mathbf{FCat}(\mathbf{Q}(\mathcal{V}))$. Every formal functor has an actual equivariant bicomodule map as its unary component.*

Proof. Corollary 2.15 identifies the unary component with an actual map. Unary substitution composes functors: pasting their two diagrams (3.2) proves the same laws for the composite. Identities and associativity are inherited from the cell calculus. $\square$

Remark 3.4 (Operations are selected cells). These definitions select cells in the constructed collections. In particular, multiplication is a finite expression in balanced cells and the inverse comparisons of Section 2; it is not an additional generator freely adjoined to a given carrier.

Definition 3.5 (Stable cotensors). For bicomodules $M : C \nrightarrow D$ and $N : D \nrightarrow E$, a cotensor cone is an equalizer with its universal factorization of balanced maps:

$$\begin{array}{ccccc} T & \xleftarrow{i} & M \otimes N & \xrightarrow[\text{1} \otimes \lambda_N]{\rho_M \otimes 1} & M \otimes D \otimes N \\ & \nwarrow \exists! \bar{\phi} & \uparrow \phi & & \\ & & Z & & \end{array} \quad (3.3)$$

Call this cone *stable* if every $K \otimes - \otimes L$ preserves it, and write its vertex as $T = M \square_D N$.

Lemma 3.6 (Stable cones represent paths). *A stable cotensor cone has unique exterior coactions making it an equivariant bicomodule cone. Its cell $(M, N) \Rightarrow T$ in $\mathbb{C}(\mathcal{V})$ is contextually universal.*

Proof. The left coaction is the unique factor in

$$\begin{array}{ccc} T & \xrightarrow{i} & M \otimes N \\ \exists! \lambda_T \downarrow & & \downarrow \lambda_M \otimes 1 \\ C \otimes T & \xrightarrow[\text{1}_C \otimes i]{} & C \otimes M \otimes N. \end{array}$$

Testing coaction equations after these monic tensors of i proves the bicomodule laws. A balanced map in any context factors uniquely through the corresponding tensor of i ; testing its remaining equations in the same way proves that the factor is balanced. Hence the cone is intrinsically contextually universal in $\mathbb{C}(\mathcal{V})$. $\square$

Theorem 3.7 (Cotensor recovery). *Whenever the cotensors used in the category and functor laws are stable, Definitions 3.1 and 3.2 are equivalent to the usual cotensor definitions: multiplication and unit are actual bicomodule maps $m : A \square_C A \rightarrow A$ and $u : C \rightarrow A$, with their cotensor associativity and unit laws, and formal functors are exactly the corresponding changing-base functors.*

Proof. By Lemma 3.6, after reversal a stable cone's cell $\iota : T \Rightarrow (M, N)$ has an inverse $\tau : (M, N) \Rightarrow T$ in the merge realization. Precomposition with τ is a single-wire substitution along T , retained in $\mathbf{Q}(\mathcal{V})$. Consequently, whenever the relevant cotensors are stable, the category operations correspond exactly to actual maps through

$$\begin{array}{ccc} (A, A) & \xrightarrow[\sim]{\tau_{A,A}} & A \square_C A \\ & \searrow \mu & \downarrow m \\ & & A \end{array} \quad \begin{array}{ccc} \emptyset_C & \xrightarrow[\sim]{\tau_0} & C \\ & \searrow \eta & \downarrow u \\ & & A, \end{array}$$

where $\tau_0 : \emptyset_C \Rightarrow C$ is the represented-unit comparison. Invert the top comparisons to recover m and u in the merge realization; unary conservativity identifies them with actual maps.

Iterated stable cotensors represent the same balanced cone functor. Unique comparisons between them give the associator and unitors, and their coherence follows because each route has the same composite to the unparenthesized tensor. Thus (3.1) becomes precisely cotensor associativity and unitality. For equivariant maps F, G over $(f, g), (g, h)$, equalizer factorization gives

$$\begin{array}{ccc} M \square_D N & \xhookrightarrow{i} & M \otimes N \\ \exists! F \square_g G \downarrow & & \downarrow F \otimes G \\ M' \square_{D'} N' & \xhookrightarrow{i'} & M' \otimes N'. \end{array}$$

Cone uniqueness proves functoriality. Transporting the diagrams (3.2) along the comparisons gives the usual unit and multiplication diagrams for cotensor functors. $\square$

Corollary 3.8 (Regular bases and the unit base). *For regular $\mathcal{V}$, formal internal categories and functors recover Aguiar's internal categories and changing-base functors. Over the unit comonoid, formal internal categories and functors are ordinary monoids and monoid maps in every monoidal $\mathcal{V}$.*

Proof. Regularity supplies the stable cotensors required by Theorem 3.7. Over I , the coactions are the unit isomorphisms, the two balance maps agree, and their equalizer is the identity of the ordinary tensor product. This cone is always stable. $\square$

4 Distributors and their restrictions

Definition 4.1 (Distributors on bicomodules). For formal categories $\mathbf{A} = (C, A)$ and $\mathbf{B} = (D, B)$, a distributor on an actual bicomodule $M : C \nrightarrow D$ has cells

$$p : (A, M) \Rightarrow M, \quad q : (M, B) \Rightarrow M$$

satisfying the two associativity squares

$$\begin{array}{ccc} (A, A, M) & \xrightarrow{(\mu_A, 1)} & (A, M) \\ (1, p) \downarrow & & \downarrow p \\ (A, M) & \xrightarrow{p} & M \end{array} \quad \begin{array}{ccc} (M, B, B) & \xrightarrow{(q, 1)} & (M, B) \\ (1, \mu_B) \downarrow & & \downarrow q \\ (M, B) & \xrightarrow{q} & M, \end{array} \quad (4.1)$$

the unit triangles

$$\begin{array}{ccc} (A, M) & \xleftarrow{(\eta_A, 1)} & M \\ & \searrow p & \parallel \\ & & M \\ & \swarrow q & \nwarrow (1, \eta_B) \\ (M, B) & & M \end{array} \quad (4.2)$$

and the commuting-action square

$$\begin{array}{ccc}
 (A, M, B) & \xrightarrow{(p,1)} & (M, B) \\
 (1,q) \downarrow & & \downarrow q \\
 (A, M) & \xrightarrow{p} & M.
 \end{array} \tag{4.3}$$

These are tree equations in $\mathbb{Q}(\mathcal{V})$. Equivariant maps over formal functors $\mathbf{F} = (f, F)$, $\mathbf{G} = (g, G)$ are actual bicomodule maps $\theta : M \rightarrow M'$, with target category carriers A', B' , making

$$\begin{array}{ccc}
 (A, M) & \xrightarrow{p} & M \\
 (F, \theta) \downarrow & & \downarrow \theta \\
 (A', M') & \xrightarrow{p'} & M'
 \end{array}
 \quad
 \begin{array}{ccc}
 (M, B) & \xrightarrow{q} & M \\
 (\theta, G) \downarrow & & \downarrow \theta \\
 (M', B') & \xrightarrow{q'} & M'.
 \end{array}$$

commute.

Proposition 4.2 (Recovery of cotensor distributors). *When the relevant category and action cotensors are stable, Definition 4.1 recovers the usual cotensor bimodules and their equivariant maps.*

Proof. Use Lemma 3.6 to identify each represented action boundary with its cotensor. Corollary 2.15 then identifies the actions with actual maps. The same coherent comparisons used in Theorem 3.7 carry the action, unit, commutation, and equivariance diagrams to their cotensor forms, and back. $\square$

Definition 4.3 (Path carriers). Let $\mathbb{K}(\mathcal{V})$ have one horizontal arrow for each path of $\mathbb{B}(\mathcal{V})$ and

$$\mathbb{K}(\mathcal{V})_{f,g}(H_1, \dots, H_n; K) = \mathbb{B}(\mathcal{V})_{f,g}(H_1 \cdots H_n; K).$$

Substitution is merge evaluation in $\mathbb{B}(\mathcal{V})$.

Proposition 4.4 (The carrier equipment). *The virtual double category $\mathbb{K}(\mathcal{V})$ is a virtual equipment. Concatenation represents lists of carriers, the empty path is its unit, and restrictions are obtained by companion and conjoint transport.*

Proof. The substitution laws, concatenation comparisons, and units come from the strict double category $\mathbb{B}(\mathcal{V})$ of Proposition 2.12. With companion and conjoint notation now taken in the *reversed* realization, boundary bending makes

$$K[f, g] = f_* K g^*$$

a cartesian restriction, with its standard square

$$\begin{array}{ccc}
 X & \xrightarrow{K[f,g]} & Y \\
 f \downarrow & \Downarrow \chi & \downarrow g \\
 X' & \xrightarrow{K} & Y'
 \end{array}$$

Its factorization property for further exterior maps follows by combining successive companion and conjoint comparisons. This gives all required cartesian restrictions. $\square$

Definition 4.5 (Formal distributors and balanced multicells). A *formal distributor* is a carrier path H_M with (4.1)–(4.3), evaluated in $\mathbb{K}(\mathcal{V})$. On a one-bicomodule carrier this is exactly the preceding definition. A multicell $\alpha : (M_1, \dots, M_n) \Longrightarrow N$ over (F, G) is a cell on the carrier paths satisfying the exterior action squares

$$\begin{array}{ccc} (A, \Gamma) & \xrightarrow{(p_{M_1, 1, \dots, 1})} & \Gamma \\ (F, \alpha) \downarrow & & \downarrow \alpha \\ (A', N) & \xrightarrow{p_N} & N \end{array} \quad \begin{array}{ccc} (\Gamma, B) & \xrightarrow{(1, \dots, 1, q_{M_n})} & \Gamma \\ (\alpha, G) \downarrow & & \downarrow \alpha \\ (N, B') & \xrightarrow{q_N} & N. \end{array}$$

Here $\Gamma = (M_1, \dots, M_n)$, with carrier paths understood, and A, B and A', B' are the source and target category carriers. For an internal interface, let Γ_i^+ be Γ with the intermediate category carrier inserted between M_i, M_{i+1} . Let r_i apply q_{M_i} and ℓ_i apply $p_{M_{i+1}}$. The internal balance square is the first diagram below. For $n = 0$, the functors have a common domain A , and the second diagram is the required nullary balance:

$$\begin{array}{ccc} \Gamma_i^+ & \xrightarrow{r_i} & \Gamma \\ \ell_i \downarrow & & \downarrow \alpha \\ \Gamma & \xrightarrow{\alpha} & N \end{array} \quad \begin{array}{ccc} A & \xrightarrow{(F, \alpha)} & (A', N) \\ (\alpha, G) \downarrow & & \downarrow p_N \\ (N, B') & \xrightarrow{q_N} & N. \end{array}$$

Proposition 4.6 (The distributor virtual double category). *Formal categories, formal functors, and the distributors and multicells of Definition 4.5 form the virtual double category*

$$\mathbb{D}_{\text{dist}}(\mathcal{V}) = \text{Mod}(\mathbb{K}(\mathcal{V}))|_{\text{FCat}(\mathcal{Q}(\mathcal{V}))}.$$

The restriction is on objects: keep the monoids whose carriers are original atomic bicomodules, and keep all distributors between them. Its vertical category is $\text{FCat}(\mathcal{Q}(\mathcal{V}))$.

Proof. The monoids and monoid maps on these carriers are exactly Definitions 3.1 and 3.2. The module equations are those of Definition 4.5, as in [2, Definition 2.8]. Substitution preserves the displayed equations by moving an inserted action through its block or across an adjacent interface. Empty blocks use the nullary equation. Identities and substitution laws are inherited from $\mathbb{K}(\mathcal{V})$. $\square$

Theorem 4.7 (Distributor restrictions and units). *The virtual double category $\mathbb{D}_{\text{dist}}(\mathcal{V})$ is a virtual equipment. Every distributor has restrictions along all formal functors. The regular distributor (A, μ_A, μ_A) is the unit at A , with universal cell η_A .*

Proof. Given N and formal functors $F = (f, F), G = (g, G)$ into its endpoints, let $\chi : H = f_* H_N g^* \Longrightarrow H_N$ be the ambient cartesian cell. Define actions by its unique factorizations below, writing N for its carrier H_N :

$$\begin{array}{ccc} (A, H) & \overset{\exists! p_H}{\dashrightarrow} & H \\ (F, \chi) \downarrow & & \downarrow \chi \\ (A', N) & \xrightarrow{p_N} & N \end{array} \quad \begin{array}{ccc} (H, B) & \overset{\exists! q_H}{\dashrightarrow} & H \\ (\chi, G) \downarrow & & \downarrow \chi \\ (N, B') & \xrightarrow{q_N} & N. \end{array} \tag{4.4}$$

The following pasting tests associativity after postcomposition with χ :

$$\begin{array}{ccccc}
(A, A, H) & \xrightarrow{(\mu_A, 1)} & (A, H) & \xrightarrow{p_H} & H \\
(F, F, \chi) \downarrow & & \downarrow (F, \chi) & & \downarrow \chi \\
(A', A', N) & \xrightarrow{(\mu_{A'}, 1)} & (A', N) & \xrightarrow{p_N} & N \\
(1, p_N) \downarrow & & \downarrow p_N & & \parallel \\
(A', N) & \xrightarrow{p_N} & N & \xlongequal{\quad} & N.
\end{array}$$

Using (4.4) on the left route identifies the two outer composites with the images under χ of the desired action composites. Cartesian uniqueness makes them equal. The unit, right action, and commuting equations follow by the same factorization argument. A factor of a balanced multicell through χ is balanced because postcomposition with χ reflects its defining equations. Thus χ is cartesian also for distributor multicells.

The regular distributor (A, μ_A, μ_A) is the unit at A , with universal cell η_A . Its contextual factorization inserts an adjacent action; balance identifies the insertion formulas and the action unit laws give their inverse. In the empty context the formula is $p_N(F, \alpha) = q_N(\alpha, G)$. Consequently $\mathbb{D}_{\text{dist}}(\mathcal{V})$ is a virtual equipment, with restrictions and units constructed above. $\square$

Corollary 4.8 (The two-sided distributor calculus). *Define*

$$\text{Dist}_{\text{poly}}(\mathcal{V}) = U_{\text{poly}} \mathbf{E}(\mathbb{D}_{\text{dist}}(\mathcal{V})).$$

There is no reversal at this stage. Its atomic arrows are individual distributors, each of which has its own bicomodule-path carrier. It preserves and reflects all balanced single-output distributor multicells, and formal functors have companions and conjoinants obtained by restricting regular distributors.

Proof. Apply Theorem 2.13 to the virtual equipment supplied by Theorem 4.7, and take the poly-virtual reduct without reversal. Lemma 2.10 supplies the companions and conjoinants. This second localization uses distributor balancing; its representing comparisons belong to that new calculus. $\square$

5 Recovery in a nonregular base

Proposition 5.1 (Ambient nonregularity). *The category $(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$ is not regular in Aguiar's sense: tensoring with $\mathbb{Z}/2$ can fail to preserve a coreflexive equalizer.*

Proof. The coreflexive pair

$$r, s : \mathbb{Z} \rightrightarrows \mathbb{Z} \oplus \mathbb{Z}, \quad r(n) = (n, 2n), \quad s(n) = (n, 0)$$

has equalizer zero. Put $R = \mathbb{Z}/2$. After tensoring, $\bar{r} = \bar{s}$, so the comparison from the tensored cone to the new equalizer is

$$\begin{array}{ccccc}
0 & \longrightarrow & R & \xrightleftharpoons[\bar{s}]{\bar{r}} & R \oplus R \\
\kappa \downarrow & & \parallel & & \parallel \\
R & \xlongequal{\quad} & R & \xrightleftharpoons[\bar{s}]{\bar{r}} & R \oplus R.
\end{array}$$

The bottom row is an equalizer and κ is not an isomorphism. Projection onto the first coordinate is a common retraction of r, s ; even this coreflexive equalizer is not preserved. $\square$

Proposition 5.2 (Stable matrix cotensors). *For a finite set S , let*

$$C_S = \bigoplus_{s \in S} \mathbb{Z}e_s, \quad \delta(e_s) = e_s \otimes e_s, \quad \varepsilon(e_s) = 1$$

be its group-like comonoid. Bicomodules $C_S \rightharpoonup C_T$ are matrices (M_{st}) of abelian groups. Their cotensors are stable and given by

$$(M \square_{C_T} N)_{su} = \bigoplus_{t \in T} M_{st} \otimes N_{tu}. \quad (5.1)$$

Proof. The coordinate endomorphisms of a coaction are orthogonal idempotents summing to the identity. Commutation of the two coactions gives the matrix decomposition; conversely, the matrix entries define the two coactions. The two balance maps insert the two intermediate basis elements. On matching indices they agree; on mismatching indices a coordinate projection of their difference is the identity. This calculation survives arbitrary tensoring, so these equalizer cones are stable, with no flatness requirement on the entries. $\square$

Corollary 5.3 (Enriched categories). *Formal categories over finite group-like comonoids, and formal functors between them, recover finite-object $\mathbf{Ab}$ -enriched categories and enriched functors.*

Proof. Proposition 5.2 and Theorem 3.7 identify the component maps as $A_{st} \otimes A_{tu} \rightarrow A_{su}$ and $\mathbb{Z} \rightarrow A_{ss}$. Comonoid maps between the C_S 's are functions of indexing sets: a group-like element over $\mathbb{Z}$ has exactly one coefficient 1 and all others 0. Thus formal functors recover enriched functors. $\square$

Proposition 5.4 (Enriched distributors and composition). *Apply the distributor construction to the sector whose carrier paths have only finite group-like bases. Its formal distributors are, up to their representing comparisons, the usual enriched distributors. Distributor composition is represented by the relative tensor product*

$$(M \otimes_{\mathbf{B}} N)_{su} = \frac{\bigoplus_t M_{st} \otimes N_{tu}}{\langle (mb) \otimes n - m \otimes (bn) \rangle}.$$

Here $b \in B_{t't'}$, $m \in M_{st}$ and $n \in N_{t'u}$, so the two terms lie in the t' and t summands.

Proof. Every carrier path has the stable matrix representative of Proposition 5.2. Transporting its actions along that comparison gives exactly an enriched distributor, by Proposition 4.2. Tensor products and finite direct sums preserve the displayed action coequalizer. More explicitly, write $X = \bigoplus_{t,t'} M_{st} \otimes B_{tt'} \otimes N_{t'u}$, $Y = \bigoplus_t M_{st} \otimes N_{tu}$ and $Q = (M \otimes_{\mathbf{B}} N)_{su}$. Balanced maps have the factorization

$$\begin{array}{ccccc} X & \xrightarrow[1 \otimes p_N]{q_M \otimes 1} & Y & \xrightarrow{\pi} & Q \\ & & \searrow \alpha & & \downarrow \exists! \bar{\alpha} \\ & & & & Z. \end{array}$$

The same diagram works in every tensor context. Hence the quotient has the contextual factorization property for balanced distributor cells. Iterated representatives are coherent by uniqueness. $\square$

Remark 5.5 (What this example establishes). Proposition 5.1 concerns the ambient category; Proposition 5.2 shows that all cotensors in this sector are stable. The next example instead places the failure of stability inside a distributor action boundary.

6 A distributor with an unrepresented action boundary

The construction has two steps. First form a category whose own composition paths are stable. Then restrict a stable distributor action along a formal functor. The resulting action remains a well-defined cell, while its input path loses contextual representability.

6.1 The category and the functor

Definition 6.1 (Triangular data). Let $D = \mathbb{Z}g \oplus \mathbb{Z}x$ with

$$\delta(g) = g \otimes g, \quad \delta(x) = g \otimes x + x \otimes g, \quad \varepsilon(g) = 1, \quad \varepsilon(x) = 0.$$

Set $E = \mathbb{Z}e \oplus D$, where e is group-like. Let $H = \mathbb{Z}a \oplus \mathbb{Z}b$ be the E - E bicomodule

$$\lambda_H(h) = e \otimes h, \quad \rho_H(a) = a \otimes g, \quad \rho_H(b) = b \otimes g + 2a \otimes x. \quad (6.1)$$

Also let $J = \mathbb{Z}t$ with $\lambda_J(t) = e \otimes t$ and $\rho_J(t) = t \otimes g$, and define $\varphi : H \rightarrow J$ by $\varphi(a) = 0$, $\varphi(b) = t$.

Lemma 6.2 (Validity of the triangular data). *The displayed formulas make E a comonoid, H, J bicomodules over E , and φ an equivariant bicomodule map.*

Proof. The comonoid laws for E hold on e, g, x : the two iterated coproducts of x are the sum of the three tensors with one x and two g 's, and the counit laws follow from $\varepsilon(g) = 1$, $\varepsilon(x) = 0$. For the right coaction on H , write $\rho_H(h) = h \otimes g + d(h) \otimes x$ with $d(a) = 0$, $d(b) = 2a$. The equation $d^2 = 0$ proves coassociativity, the first square below. Counitality and commuting coactions follow from the formulas; the laws for J use that e, g are group-like. Right equivariance of φ is the second square, since $\varphi d = 0$; left equivariance follows from the common support e :

$$\begin{array}{ccc} H & \xrightarrow{\rho_H} & H \otimes E \\ \rho_H \downarrow & & \downarrow 1 \otimes \delta_E \\ H \otimes E & \xrightarrow{\rho_H \otimes 1} & H \otimes E \otimes E \end{array} \quad \begin{array}{ccc} H & \xrightarrow{\varphi} & J \\ \rho_H \downarrow & & \downarrow \rho_J \\ H \otimes E & \xrightarrow{\varphi \otimes 1} & J \otimes E. \end{array}$$

□

Lemma 6.3 (Triangular categories). *For $K = H$ or J , the carrier $A_K = E \oplus K$ defines a formal category, with unit the inclusion of E and multiplication the fold*

$$m_K : E \oplus K \oplus K \longrightarrow E \oplus K, \quad (c, k_1, k_2) \longmapsto (c, k_1 + k_2).$$

Every finite composition path is stably represented; for $n \geq 1$,

$$A_K^{\square E^n} \cong E \oplus K^{\oplus n}.$$

Proof. Expand $(E \oplus K) \otimes (E \oplus K)$ into its four blocks. The E, E, E, K , and K, E blocks have the stable unit cotensors E, K , and K . The K, K block has stable equalizer zero: projection of the inserted E factor onto its e coordinate takes the two balance maps to zero and the identity, respectively. Thus every equalizing map is zero, also after tensoring.

For a longer word, remove regular E factors using their stable unit cones. A block with no K contributes E , one K contributes K , and two or more K 's contribute zero. This proves the asserted representatives and stability. Associativity is the fold square

$$\begin{array}{ccc} E \oplus K^{\oplus 3} & \xrightarrow{\nabla_{12}} & E \oplus K^{\oplus 2} \\ \nabla_{23} \downarrow & & \downarrow m_K \\ E \oplus K^{\oplus 2} & \xrightarrow{m_K} & E \oplus K. \end{array}$$

Here ∇_{12} and ∇_{23} fold the indicated pair of K summands; both routes add all three. The unit equations are the summand inclusions. Theorem 3.7 converts these into formal category laws. $\square$

Corollary 6.4 (The restriction functor). *Write $A = E \oplus H$ and $B = E \oplus J$, with the resulting categories $\mathbf{A}, \mathbf{B}$. The map*

$$F = 1_E \oplus \varphi : A \longrightarrow B$$

defines a formal functor $\mathbf{F} = (1_E, F) : \mathbf{A} \rightarrow \mathbf{B}$.

Proof. By Lemma 6.2, F is equivariant. It preserves the unit and the folds of Lemma 6.3 through

$$\begin{array}{ccc} E \oplus H^{\oplus 2} & \xrightarrow{m_H} & A \\ 1_E \oplus \varphi^{\oplus 2} \downarrow & & \downarrow F \\ E \oplus J^{\oplus 2} & \xrightarrow{m_J} & B \end{array} \quad \begin{array}{ccc} E & \xleftarrow{u_A} & A \\ & \searrow u_B & \downarrow F \\ & & B. \end{array}$$

Theorem 3.7 gives the formal functor laws. The carriers A, B, E have ranks five, four, and three, respectively. $\square$

6.2 An actual bicomodule with a nonzero formal action

Definition 6.5 (The distributor carrier). Put $N = \mathbb{Z}/4$ and $M = Nu \oplus Nv$, with

$$\lambda_M(u) = e \otimes u, \quad \lambda_M(v) = g \otimes v,$$

and the right coaction of the unit comonoid $\mathbb{Z}$. Let 1 denote the identity formal category at $\mathbb{Z}$.

Proposition 6.6 (The stable action). *The carrier M admits a $\mathbf{B}$ -1 distributor structure. Its action paths (B, M) and (B, B, M) are stably represented by $M \oplus N$ and $M \oplus N \oplus N$, respectively. Its left action is induced by the actual equivariant map*

$$p_{B,0} : M \oplus N \longrightarrow M, \quad ((ru + sv), w) \longmapsto (r + w)u + sv.$$

Proof. For (B, M) , the regular E block gives M ; $J \otimes Nu$ has mismatched supports and gives zero; and both coactions on $J \otimes Nv$ insert g , giving all of N . These computations persist under arbitrary tensoring. The extra N summand and its image Nu both have left support e , so this is equivariant. The triple (B, B, M) is stably represented by $M \oplus N \oplus N$, and the action square becomes

$$\begin{array}{ccc} M \oplus N \oplus N & \xrightarrow{1_M \oplus +} & M \oplus N \\ \sigma \downarrow & & \downarrow p_{B,0} \\ M \oplus N & \xrightarrow{p_{B,0}} & M. \end{array}$$

Here $\sigma(m, w_1, w_2) = (p_{B,0}(m, w_2), w_1)$; both routes send $((ru + sv), w_1, w_2)$ to $(r + w_1 + w_2)u + sv$. The unit selects M . Hence this gives an associative, unital formal action p_B . The right action $q : (M, \mathbb{Z}) \Rightarrow M$ is the unit comparison and commutes with p_B by its coherence. $\square$

Proposition 6.7 (Restriction on the actual carrier). *The same bicomodule M admits a $\mathbf{A}$ -1 distributor structure, obtained by restriction along $\mathbf{F}$ of Corollary 6.4. Its left action is defined by the outer route*

$$\begin{array}{ccc} (A, M) & \xrightarrow{(F, 1_M)} & (B, M) \\ p_A \downarrow & & \sim \downarrow \tau_{B,M} \\ M & \xleftarrow{p_{B,0}} & M \oplus N \end{array} \quad (6.2)$$

where $\tau_{B,M}$ is the stable comparison already constructed. Equivalently, $p_A = p_B(F, 1_M)$, a tree substitution in $\mathbf{Q}(\mathbf{Ab})$. Its right action is the unit action q of Proposition 6.6.

Proof. Since the base map of $\mathbf{F}$ is 1_E , restriction retains the actual carrier M . Associativity follows by pasting

$$\begin{array}{ccccc}
(A, A, M) & \xrightarrow{(\mu_A, 1)} & (A, M) & \xrightarrow{p_A} & M \\
(F, F, 1) \downarrow & & \downarrow (F, 1) & & \parallel \\
(B, B, M) & \xrightarrow{(\mu_B, 1)} & (B, M) & \xrightarrow{p_B} & M \\
(1, p_B) \downarrow & & \downarrow p_B & & \parallel \\
(B, M) & \xrightarrow{p_B} & M & \xlongequal{\quad} & M.
\end{array}$$

The two routes around the outside are the two A -action composites, after using the definition of p_A on the left route. Unitality is the left triangle in

$$\begin{array}{ccccc}
M & \xrightarrow{(\eta_A, 1)} & (A, M) & \xrightarrow{(F, 1)} & (B, M) \\
& \searrow & \downarrow p_A & \swarrow p_B & \\
& & M & &
\end{array}$$

whose outer triangle is the functor unit law followed by the B -action unit law. The commuting-action equation follows by substituting F into that for p_B, q . Thus (M, p_A, q) is a formal distributor $A \nrightarrow 1$ on one actual bicomodule. The construction of p_A uses the inverse comparison for the stable path (B, M) . $\square$

6.3 The unstable cone and the contextual obstruction

Lemma 6.8 (The unstable action cotensor). *The ordinary equalizer $T = A \square_E M$ exists and carries an equivariant $E\text{-}\mathbb{Z}$ bicomodule cone. After tensoring with $R = \mathbb{Z}/2$, the comparison from this cone to the new equalizer is neither monic nor epic. In particular, the action cotensor is not stable.*

Proof. The regular block of T is M . The $H \otimes Nu$ block is zero by its supports, while an element

$$a \otimes \alpha v + b \otimes \beta v \in H \otimes Nv$$

is balanced exactly when $2\beta = 0$ in N . Indeed its balance obstruction is $2a \otimes x \otimes \beta v$. Consequently

$$T \cong M \oplus Na \oplus N[2]b \cong (\mathbb{Z}/4)^3 \oplus \mathbb{Z}/2. \quad (6.3)$$

Writing z for the last, order-two generator, its cone $i : T \hookrightarrow A \otimes M$ is

$$u \mapsto e \otimes u, \quad v \mapsto g \otimes v, \quad a \mapsto a \otimes v, \quad z \mapsto 2b \otimes v. \quad (6.4)$$

Give u, a, z left support e , and v left support g ; this makes T an $E\text{-}\mathbb{Z}$ bicomodule and i an equivariant cone.

After tensoring with $R = \mathbb{Z}/2$, the nonzero class $z \otimes \bar{1}$ maps to zero. Meanwhile

$$b \otimes v \otimes \bar{1} \quad (6.5)$$

becomes balanced: its obstruction is a multiple of 2. This element cannot lie in the image of $i \otimes 1_R$, whose b coordinate is zero. Thus the tensored cone fails both injectivity and surjectivity onto the new equalizer T_R^{bal} . The comparison is

$$\begin{array}{ccc}
T \otimes R & \xrightarrow{i \otimes 1_R} & A \otimes M \otimes R \\
\kappa \downarrow & & \parallel \\
T_R^{\text{bal}} & \xrightarrow{i_R} & A \otimes M \otimes R.
\end{array}$$

Both objects on the left have underlying group $(\mathbb{Z}/2)^4$. In the source basis (u, v, a, z) and the target basis given by $e \otimes u, g \otimes v, a \otimes v, b \otimes v$ after reduction, κ has matrix $\text{diag}(1, 1, 1, 0)$. The failure is in this canonical cone, whose image has dimension three. $\square$

Theorem 6.9 (An unrepresented action boundary). *The action boundary (A, M) has no atomic contextual representative. Its formal action operates nontrivially on the new balanced element (6.5).*

Proof. In the original globular calculus P of $\mathbb{C}(\mathbf{Ab})$, the cone i is a cell $(A, M) \Rightarrow T$. It represents the empty-context balanced-map functor: factor a balanced map through its ordinary equalizer, and test equivariance of the factor after $1_E \otimes i$. This map is monic because E is free. The right coaction is the unit coaction.

Let $W = \mathbb{Z}/2$ have left E -coaction $w \mapsto e \otimes w$ and right unit coaction. With R regarded as a $\mathbb{Z}$ - $\mathbb{Z}$ bicomodule, the map

$$\beta : W \longrightarrow A \otimes M \otimes R, \quad \bar{1} \longmapsto b \otimes v \otimes \bar{1}$$

satisfies both interior balances and exterior equivariance. It is a cell in $P(A, M, R; W)$ for which the factorization triangle fails:

$$\begin{array}{ccc} W & \xrightarrow{\beta} & A \otimes M \otimes R \\ \downarrow \#h & \nearrow i \otimes 1_R & \\ T \otimes R & & \end{array}$$

The dashed arrow denotes the nonexistent factor: the b coordinate of $\beta(\bar{1})$ is nonzero, while that of every element in the image of $i \otimes 1_R$ vanishes. Hence i fails the contextual test of Definition 2.5.

Every other contextual representative would first represent the same empty-context functor, so its cone would be i composed with a bicomodule isomorphism:

$$\begin{array}{ccc} S & \xrightarrow[\sim]{\theta} & T \\ & \searrow j & \downarrow i \\ & & A \otimes M. \end{array}$$

Here j is the proposed representing cone. Tensoring θ with R cannot change this failure of factorization. No alternative atom can therefore represent the path. By Theorem 2.8, (A, M) is not isomorphic to any atom in the globular localization, or in its reversed merge realization.

In the forward realization, β is the primitive cell $W \Rightarrow (A, M, R)$. Applying F to its first output replaces b by t . The stable comparison for (B, M) followed by p_B then sends $t \otimes v \otimes \bar{1}$ to $u \otimes \bar{1}$. Thus, in the merge realization, the entire evaluation is

$$\begin{array}{ccccc} W & \xrightarrow{\beta} & (A, M, R) & \xrightarrow{(F, 1, 1)} & (B, M, R) \\ \downarrow \gamma & & \downarrow p_A \odot 1_R & & \downarrow \sim \tau_{B, M} \odot 1_R \\ (M, R) & \xlongequal{\quad} & (M, R) & \xleftarrow[p_{B, 0} \odot 1_R]{} & (M \oplus N, R) \end{array}$$

Here (M, R) is stably represented by $M \otimes R$. Under this comparison, unary conservativity identifies γ with the actual map $\bar{1} \mapsto u \otimes \bar{1} \neq 0$. The reduct retains the resulting cells and equality. $\square$

Corollary 6.10 (The action on the ordinary cotensor). *The map $p_T : T \rightarrow M$ induced by the formal action is nonzero on the order-two summand of T : it sends its generator z to $2u$.*

Proof. The primitive cone gives a triangle in the merge realization

$$\begin{array}{ccc} T & \xrightarrow{\iota} & (A, M) \\ & \searrow p_T & \downarrow p_A \\ & & M, \end{array}$$

where unary conservativity identifies p_T with the actual map

$$p_T : T \longrightarrow M, \quad (ru + sv, \alpha, \beta) \longmapsto (r + \beta)u + sv, \quad \beta \in N[2],$$

so $p_T(z) = 2u \neq 0$. Nevertheless, the contextual action above cannot be evaluated by first factoring through $T \otimes R$: that factorization does not exist by Theorem 6.9. $\square$

Corollary 6.11 (Restriction does not preserve action stability). *Stability of action cotensors need not survive restriction along a formal functor, even when its base map is the identity and both formal categories have stable finite composition paths. This failure occurs for a distributor on one actual bicomodule, whose restricted action boundary has no contextual atomic representative.*

Proof. Use the functor $F : \mathbf{A} \rightarrow \mathbf{B}$ of Corollary 6.4. Lemma 6.3 gives stable composition paths for both categories. The $\mathbf{B}$ -action of Proposition 6.6 has a stable action cotensor; its restriction is the $\mathbf{A}$ -action of Proposition 6.7. Lemma 6.8 and Theorem 6.9 give the asserted instability and nonrepresentability. $\square$

Remark 6.12 (A prime-indexed family). Replacing $2, 4$ by p, p^2 gives the same construction for every prime p : use $\rho_H(b) = b \otimes g + pa \otimes x$, $N = \mathbb{Z}/p^2$, and $R = \mathbb{Z}/p$. The problematic summand is $N[p]b$, its generator has cone image $pb \otimes v$, and the reduced comparison again has matrix $\text{diag}(1, 1, 1, 0)$. The proofs above apply with these substitutions.

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